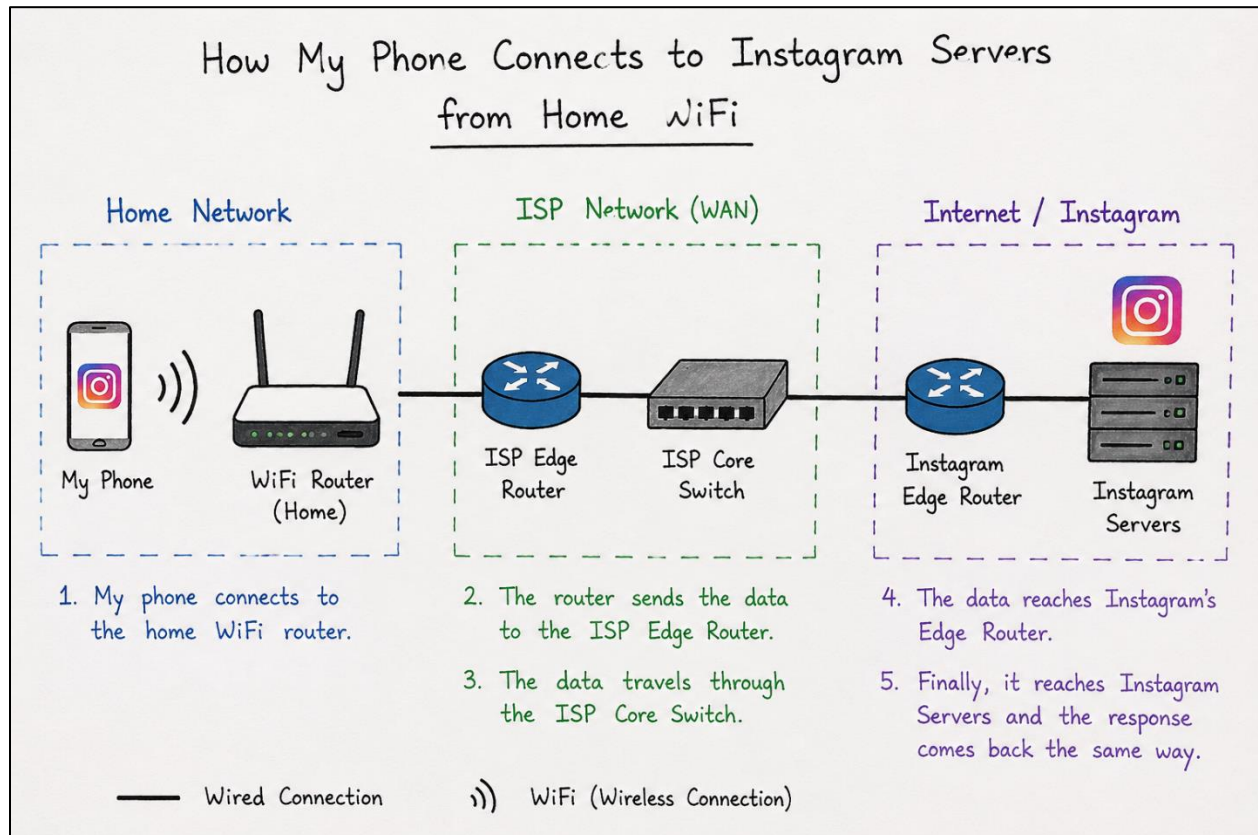


1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

In simple terms:

- **MPLS:** Reliable private business network.
- **Leased Line:** Dedicated connection with consistent performance.
- **Broadband:** Affordable Internet connection for everyday use.

WAN Technology	Speed	Cost	Real-Life Use
MPLS (Multiprotocol Label Switching)	High and reliable; suitable for business-critical traffic	High	Large companies such as a Zomato office connecting multiple branches/data centers
Leased Line	High, dedicated and consistent speed	Very High	Organizations such as IRCTC or banks that need a dedicated connection between locations
Broadband Internet	Medium to high; speed can vary depending on provider and network traffic	Low	Home users , small offices, and students accessing websites and apps

3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

How a VPN Uses a WAN on Public Wi-Fi

- A **VPN (Virtual Private Network)** creates a secure, encrypted connection between your device and a VPN server over a **WAN (Wide Area Network)** such as the Internet.

Example: Listening to Spotify at a coffee shop

1. **Connect to public Wi-Fi:** Your phone/laptop connects to the coffee shop's Wi-Fi router.
2. **Connect to the WAN:** The Wi-Fi router sends your Internet traffic through the coffee shop's ISP and the wider Internet (WAN).
3. **VPN encryption:** The VPN app encrypts your traffic and sends it through a secure
4. **VPN tunnel** to a VPN server.
5. **VPN server → Spotify:** The VPN server forwards your request to Spotify's servers and sends the response back through the encrypted tunnel.
6. **Data reaches your device:** Your VPN decrypts the returned data so the Spotify app can use it.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Recommended WAN Technology: MPLS

- For Flipkart connecting warehouses in **Ahmedabad, Mumbai, and Delhi**, I would recommend **MPLS (Multiprotocol Label Switching)**.

Two reasons:

1. **Reliable and consistent connectivity:** MPLS provides predictable performance between warehouses, which is useful for inventory, order-processing, and internal business applications.
2. **Secure and scalable:** MPLS can connect multiple locations through a managed private WAN and can be expanded as Flipkart adds more warehouses.